PROBLEM 10

Conservative Force and Potential I

Original statement · Problems.pdf, p. 9

Work & Potential Energy

A particle of mass m moves under a force $\mathbf{F} = -cx^3\mathbf{i}$, where c is a positive constant. Find the potential energy function. If the particle starts from rest at $x = -a$, what is its velocity when it reaches $x = 0$? In the subsequent motion, where does it stop again?

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m moves along the Ox axis under the force $\mathbf{F} = -cx^3\,\mathbf{i}$, with $c > 0$, a restoring force toward the origin.
- **Constraints:** the particle is confined to the axis, $y = z = 0$ (holonomic, bilateral, smooth). The guide exerts reactions perpendicular to Ox, which do not enter the motion along x.
- **Degrees of freedom:** $\mathrm{CDOF} = 3 - 2 = 1$, with the coordinate $x(t)$.
- **Initial conditions:** released from rest at $x(0) = -a$ ($a > 0$): $\dot{x}(0) = 0$.
- **Parameters:** m [kg]; c [N/m^3 = kg m^{-2} s^{-2}]; a [m].

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, fixed Cartesian, with $\mathbf{i}$ along the motion. The unit vectors are constant, so no basis change is needed.

Derivation

3.1 Kinematics

Pedagogical Justification: a single Cartesian coordinate describes the motion, and its unit vector is constant.

$$\mathbf{r} = x\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i}$$

3.2 Forces

- **Applied force:** $\mathbf{F} = -cx^3\,\mathbf{i}$. **Constraint forces:** the reaction of the guide, perpendicular to $\mathbf{i}$. **Inertia forces:** none (S_0 is inertial).

Pedagogical Justification: the force depends only on x, so $F_x\,dx$ is the exact differential of a function of x and the force is conservative (Notes Sec. 4.4). The potential energy follows from the work, with the reference $V(0) = 0$ and Barrow's rule:

$$V(x) - V(0) = -\int_0^x F_x(x')\,dx' = -\int_0^x \left(-cx'^3\right) dx' = c\left[\frac{x'^4}{4}\right]_0^x = \frac{1}{4}cx^4 - 0 \implies V(x) = \frac{1}{4}cx^4$$

3.3 Equations of Motion

- **Newton's second law along i:** $m\ddot{x} = -cx^3$. One unknown, $x(t)$, and one equation.
- **First integral.** Multiplying by $\dot{x}$ and using the chain rule $\frac{d}{dt}\left(\frac{1}{2}\dot{x}^2\right) = \dot{x}\ddot{x}$ and $\frac{d}{dt}V(x) = V'(x)\dot{x} = cx^3\dot{x}$:

$$\frac{d}{dt}\left(\tfrac{1}{2}m\dot{x}^2 + V(x)\right) = \dot{x}\left(m\ddot{x} + cx^3\right) = 0 \implies E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{4}cx^4 = \text{const}$$

3.4 Integration and Answers

Total energy

- The energy is evaluated at the release instant ($x = -a$, $\dot{x} = 0$):

$$E = \tfrac{1}{2}m \cdot 0 + \tfrac{1}{4}c(-a)^4 = \tfrac{1}{4}ca^4$$

Velocity at $x = 0$

> **Pedagogical Justification:** energy conservation between the release point and $x = 0$, where $V(0) = 0$:

$$\tfrac{1}{4}ca^4 = \tfrac{1}{2}mv^2 + 0 \implies v^2 = \frac{ca^4}{2m} \implies |v| = a^2\sqrt{\frac{c}{2m}}$$

- **Direction.** For $x < 0$ the force $F_x = -cx^3 = c|x|^3 > 0$ points along $+\mathbf{i}$, so the particle accelerates toward the right and crosses $x = 0$ with positive velocity:

$$\boxed{\mathbf{v}(0) = a^2\sqrt{\frac{c}{2m}}\,\mathbf{i}}$$

Where it stops again

> **Pedagogical Justification:** the particle stops when $\dot{x} = 0$, i.e. when the kinetic energy vanishes, so $V(x_{\text{stop}}) = E$:

$$\tfrac{1}{4}cx_{\text{stop}}^4 = \tfrac{1}{4}ca^4 \implies x_{\text{stop}}^4 - a^4 = \left(x_{\text{stop}}^2 - a^2\right)\left(x_{\text{stop}}^2 + a^2\right) = (x_{\text{stop}} - a)\,(x_{\text{stop}} + a)\left(x_{\text{stop}}^2 + a^2\right) = 0$$

The last factor is positive for real x, so $x_{\text{stop}} = \pm a$. Since the particle started at $-a$ and has crossed the origin moving toward $+x$, the next stop is

$$\boxed{x_{\text{stop}} = +a}$$

Interpretation & Checks

- **Symmetry and anharmonic oscillation:** $V(x) = \frac{1}{4}cx^4$ is even and convex ($V'' = 3cx^2 \geq 0$), so the particle oscillates periodically between $x = -a$ and $x = +a$.
- **Dimensional check:** $[ca^4] = \mathrm{N\,m^{-3}\,m^4} = \mathrm{J}$ and $[v] = \mathrm{m^2}\sqrt{\dfrac{\mathrm{kg\,m^{-2}s^{-2}}}{\mathrm{kg}}} = \mathrm{m^2\,m^{-1}s^{-1}} = \mathrm{m/s}$.
- **Limit:** for $a \to 0$ the release energy vanishes and $v \to 0$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 10 in its solution keys. The results are verified by independent derivation and by numerical integration (DOP853, $m = c = a = 1$): $|v(0)| = 0.707107\ \mathrm{m/s} = \sqrt{1/2}$ and the successive stops are at $x = +1, -1, +1$.

Source: Problems.pdf